

Frédéric Andrian

Junior AI/ML Agent Engineer

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Professional Summary

Junior AI/ML Engineer completing an MSc-equivalent engineering degree in Artificial Intelligence and Cognitive Science. At INRIA, independently built an end-to-end multi-step LLM agent without an orchestration framework to understand and control tool execution, state, memory, context, and multi-step behavior. Developed Python and FastAPI services, REST APIs, MongoDB persistence, RAG, webhooks, Dockerized components, and GitHub CI/CD. Through software engineering projects at ENSC and personal projects, I have built complete applications and explored several technologies, including Java. I am motivated to continue learning and to build reliable backend services for production agent workflows.

Experience

INRIA (French National Institute for Research in Digital Science and Technology), Mnémosyne Team

Bordeaux, France

AI Research Engineer Intern

Mar. 2026 – Sep. 2026

- Independently designed and developed, with guidance from my supervisors, an end-to-end multi-step AI agent that helps researchers discover scientific papers.
- Built the agent without an orchestration framework to understand and control its tools, state, memory, context management, and multi-step execution mechanisms.
- Integrated the OpenAI API and designed the tools used by the agent to retrieve information and complete successive tasks.
- Built Python and FastAPI backend services, REST APIs documented with OpenAPI, and MongoDB persistence.
- Developed a React and Next.js interface that enabled researchers to interact with the agent and its tools.
- Implemented a RAG pipeline using document chunking, embeddings, semantic search, and the ChromaDB vector database.
- Developed a Python tool to automatically collect scientific paper metadata from multiple public sources.
- Implemented webhooks to relay the results of scheduled tasks.
- Evaluated OpenClaw within an isolated Docker environment, with attention to risks created by autonomous tool and file access.
- Dockerized selected tools and implemented a GitHub CI/CD pipeline to automate integration.
- Managed a Dockerized web application for LLMs installed locally on a Mac Studio, including model installation and configuration.
- Benchmarked Qwen, Gemma, and GLM on an HPC cluster with Slurm across latency, TTFT, response quality, and context size.
- Evaluated multiple quantization configurations to identify the best trade-off between inference performance and response quality for each execution environment.
- Defined an experimental protocol, analyzed the results, and contributed to a scientific manuscript documenting the model evaluation.

Karunya University

Coimbatore, India

Reinforcement Learning Research Intern

Jun. 2025 – Sep. 2025

- Designed, implemented, and compared deterministic path planning with a reinforcement learning approach for autonomous navigation of a hospital robot.
- Deployed and optimized the software and reinforcement learning model on an embedded Raspberry Pi.

Technical Skills

Backend Development Agentic AI

Python, Java, FastAPI, Node.js, REST APIs, OpenAPI, MongoDB, SQL
OpenAI API, framework-free multi-step agents, tool design and orchestration, memory and context management

Agent Frameworks RAG

LangChain experience from an academic project, OpenClaw evaluation
Chunking, embeddings, semantic search, ChromaDB

Integration and Automation Deployment

Webhooks, scheduled tasks, Git/GitHub, CI/CD
Linux, Docker, Slurm, HPC environments, Raspberry Pi

Model Evaluation

Qwen, Gemma, GLM, quantization, latency, TTFT, response quality, context size

Machine Learning Languages

PyTorch, Deep Learning, Reinforcement Learning, data preprocessing, NLP
French (native), English (professional, IELTS 7.5), Spanish (basic)

Education

ENSC, Bordeaux INP

Bordeaux, France

Engineering Degree, MSc-equivalent, in Artificial Intelligence and Cognitive Science

2023 – 2026

- Coursework in software engineering, machine learning, large language models, natural language processing, deep learning, and human-computer interaction.